

Reflective and Interrupted Light

Qualify copied optical samples without inventing hardware facts

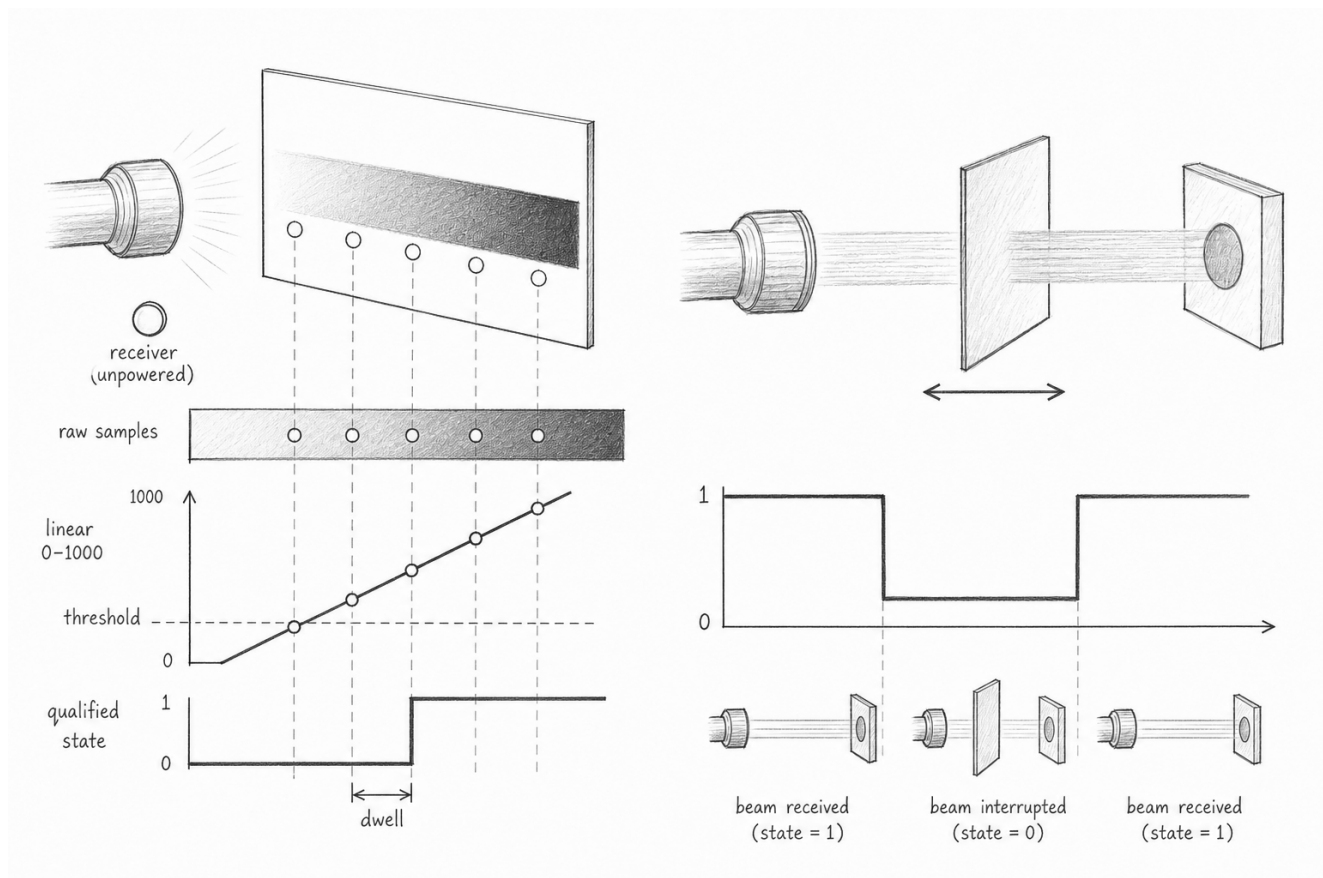
Synthetic E0 experiment; powered fixture and bench acceptance gated

Lesson	040 — component	Time	90–150 minutes
Prerequisites	Lessons 007–009	Board	host replay; Mega sketch is synthetic
Evidence	copied trace and policy snapshots	Status	E0 host integration active; bench open

Scope gate. This lesson qualifies supplied ADC counts and digital levels. It publishes no powered adapter, wiring table, pin assignment, or formal schematic. Family names do not establish an electrical identity. No lux, reflectance, distance, navigation, collision-avoidance, ambient rejection, or crosstalk claim follows from these policies.

1 Two sources, two narrow policies

`ReflectiveObservationPolicy` accepts a reflective sample; `BeamObservationPolicy` accepts a beam sample. They share provenance and quality vocabulary, but they are not a universal optical driver. Each pure policy owns copied state only: no endpoint, resource claim, emitter, interrupt, timer, callback, heap, or hidden clock.



Alternative text: a graphite worksheet maps five reflective raw samples through a straight 0–1000 normalization line, then shows a separate qualified-state step after threshold dwell. Beside it, a hand-moved card interrupts an abstract beam and produces a high/low trace for one configured polarity. The drawing is conceptual, unpowered, and not an electrical schematic.

The caller supplies **OpticalProvenance**: source ID, calibration revision, and observation time. Identity is typed. Equal numeric IDs from a reflective and interrupted source never mean that the sources are the same.

2 Normalize reflective evidence

A valid reflective configuration names a qualified ADC interval, two distinct references inside it, activation and release thresholds, dwell, and active polarity. Widened integer arithmetic maps the references onto 0–1000 and clamps beyond them:

$$N = \text{clamp} \left(\frac{1000(\text{raw} - \text{darkReference})}{\text{lightReference} - \text{darkReference}}, 0, 1000 \right).$$

The implementation handles either reference ordering without unsigned underflow. The raw count and both references remain in every observation.

Activation and release are separate hysteresis boundaries interpreted relative to **darkerIsActive**. A candidate must remain continuously on its side of the threshold for the configured dwell. The output **markerActive** is polarity-independent; later policy never needs to reconstruct the circuit’s direction.

Quality	Meaning
Unqualified	initialized or reset, with no qualified observation
Valid	source status, range, calibration, and time are acceptable
BelowQualifiedRange / AboveQualifiedRange	raw evidence is outside the configured interval; no electrical diagnosis
DegenerateCalibration	the supplied references cannot define a span
SourceFault	the copied source status or level is unusable
TimingFault	supplied timestamps violate the deterministic order

3 Qualify interrupted-light edges

The beam policy maps one configured digital level to **interrupted**. It uses independent interruption and restoration dwell times. Pulses shorter than a boundary, bounce, and chatter do not become qualified edges. An edge event lasts for one observation and clears on the next later update.

The words “beam” and “interrupted” are semantic labels for copied evidence. A digital level cannot establish an emitter’s wavelength, current, geometry, range, eye safety, pull-up, or output rail. It also cannot establish that a retail tracking or avoidance board is electrically equivalent.

4 Time before interpretation

Time validation precedes enum, source, range, and threshold processing. Identical same-time samples are idempotent. Changed same-time evidence, backward apparent time, or an exact-or-greater unsigned half-range jump publishes **TimingFault** without partially applying the changed sample. Reset clears candidates, edge events, and prior qualification.

Deterministic rule. A saved configuration, timestamp sequence, raw samples, and source statuses must reproduce every field exactly. Sampling between supplied timestamps is unknowable and must not be invented.

5 Predict, observe, interpret

1. Replay a slow darkening ramp. Predict that crossing the activation threshold alone is insufficient; the complete dwell must also pass.
2. Observe raw count, normalized value, stable duration, and the one-update activation event. Interpret the event only as calibration-bound marker evidence.
3. Reverse the ramp. Predict retention inside the hysteresis band and release only after the distinct release threshold and dwell.
4. Replay one beam pulse just shorter than interrupt dwell and one exactly equal to it. Predict suppression, then one interruption event.
5. Supply changed evidence at the same timestamp. Predict `TimingFault` with no partially updated observation.
6. Reset and replay the trace. Compare every output field, not only the final boolean state.

For this E0 lesson, the copied trace and snapshot table are the non-Serial observation path. A future powered circuit must add independent raw, qualified-event, and ready/fault signals. Initialization evidence and electrical safe-state evidence remain separate checks.

6 Fault and comparison lab

Test both reference orderings and active polarities. Exercise each qualified range edge at minus one, exact, and plus one; each threshold boundary; dwell at short, exact, and long durations; same-time replay; rollover; half-range; and reset recovery. Use paired ambient traces and adjacent-source traces only to compare recorded evidence. Correlation is not proof of ambient compensation or crosstalk.

An out-of-range ADC result cannot distinguish open, short, floating wire, surface material, distance, or illumination. A stuck digital level cannot distinguish a stationary card from a source, receiver, wiring, or adapter fault. Those diagnoses require an exact powered circuit and measured evidence.

7 Powered-work gate and open acceptance

Before any physical adapter, wiring table, formal schematic, or E1 example is published, one exact specimen must establish markings, pin order, supply, output rails, polarity, pull-ups, emitter behavior, current, and an authoritative electrical source. `Laser Emit` remains excluded and unpowered.

On a future qualified bench, record unpowered identity and continuity first; then supply and output ranges, per-path and aggregate current, raw and event test points, failure behavior, shutdown, reset, and power removal. Stop for heat, odor, reset, unstable rail, or an unexpected output. Compilation, deterministic replay, and this PDF are not physical acceptance.

8 Software evidence and next use

The deterministic tests cover configuration, ranges, normalization, hysteresis, dwell, pulse filtering, events, status precedence, time order, rollover, and replay. Lesson 041 will consume copied qualified evidence while retaining its source identity, status, age, and disagreement; it will not requalify these optical samples.

The canonical HTML lesson links the public header, sketch, governing design record, testing contract, PDF policy, and safety model. This untagged print edition is its bench companion and makes no PDF/UA or physical accessibility claim.